

GEO LATENT: GEOMETRY-GUIDED LATENT STRUCTURING WITH ROUTED OPTIMIZATION FOR 3D REASONING

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ABSTRACT

Despite progress in vision-language models, 3D spatial reasoning from 2D images remains challenging. Text-based methods describe intermediate geometry with discrete tokens, limiting fidelity for continuous spatial relations. Continuous latents offer richer representations, but a single latent type does not explicitly separate the cues needed across spatial tasks. Decomposed spatial latents address this by representing position, direction, and global geometry separately under geometric supervision. Yet the geometry representation can still collapse toward one dominant direction, and unrestricted attention can leave the latents underused during answer learning. We introduce **GeoLatent**, combining Common-Residual Geometry Alignment (CR-GEO) with routed optimization to structure the geometry states while promoting latent-mediated answer learning. CR-GEO separates shared from residual teacher geometry; routed optimization jointly trains geometry and language, temporarily directs visual answer learning through the latents, and restores full attention with geometry supervision. In controlled comparisons, CR-GEO raises geometry effective rank from 1.00 to 3.87, while blocking latent readout at the bottleneck lowers direction accuracy from 89.1% to 25.8% on 128 fixed questions. After recovery, the differentiated geometry representation and latent-mediated visual route remain available alongside direct image access. GeoLatent achieves 73.0% on SPAR-Bench and 72.1% on SPBench, outperforming previously reported methods on both.

1 INTRODUCTION

Vision-language models (VLMs) have achieved remarkable progress in visual understanding and reasoning, yet 3D spatial reasoning remains challenging (Zhang et al., 2025b; Li et al., 2026d). Spatial reasoning tasks require models to infer geometric properties such as depth, distance, relative position, orientation, and scene layout from 2D observations. To improve these capabilities, existing studies have explored spatial instruction tuning, progressive training, and explicit 3D or 2.5D representations (Chen et al., 2024; Li et al., 2026d; Wu et al., 2025a). However, when intermediate reasoning is explicitly modeled, it is still commonly expressed as text-based chain-of-thought. Continuous geometric information must then be represented through discrete language or numerical tokens, which may lose fine-grained spatial detail and make the reasoning more susceptible to language priors (Li et al., 2026c).

Continuous latent reasoning provides an alternative by representing intermediate reasoning in continuous hidden states rather than expressing every step as natural-language tokens (Shen et al., 2025; Li et al., 2026e). For spatial reasoning, however, these latent representations need to capture task-relevant spatial cues at different levels of granularity: some tasks require precise object positions or relative directions, whereas others depend on more global scene geometry. With a single latent type, spatial cues with different functions and levels of granularity have to share the same form of latent representation, making it difficult to accommodate the diverse demands of spatial reasoning tasks. GeoAnchor addresses this limitation by decomposing intermediate 3D spatial information into three

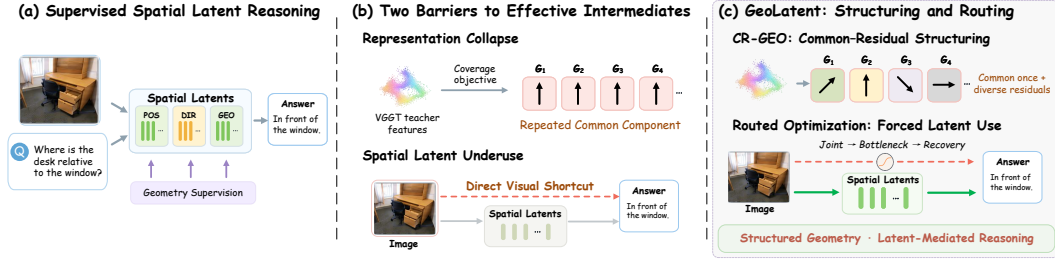


Figure 1: GeoLatent addresses two limitations of decomposed latent reasoning: CR-GEO reduces redundancy among geometry states, while routed optimization encourages answer learning to use the decomposed spatial latents.

complementary latent components: position, direction, and geometry latents, which represent object positions, relative directions, and global scene structure, respectively (Li et al., 2026c). These latent components are trained with corresponding geometric supervision. However, it remains unclear whether geometric supervision alone is sufficient to yield effective latent representations and whether these latent intermediates consistently contribute to subsequent reasoning and answer prediction.

Our analysis reveals two limitations that geometric supervision alone does not resolve. First, when the teacher geometry contains a dominant shared component, GeoAnchor’s geometry alignment objective can lead multiple geometry states to learn highly similar representations rather than capture complementary scene structure. We observe that the resulting geometry representation has an effective rank close to one, indicating substantial redundancy across its states. Second, inserting decomposed spatial latents into the reasoning trajectory does not ensure that subsequent reasoning and answer prediction actually use them. Under standard multimodal attention, downstream text and answer tokens can still directly access the original image tokens, allowing the model to answer with limited reliance on the latent intermediates. Thus, geometric supervision alone does not ensure either differentiated geometry representations or effective use of the latent intermediates during answer learning.

As summarized in Figure 1, we introduce **GeoLatent**, which improves both the geometric representations learned by decomposed spatial latents and their role in subsequent spatial reasoning. Within GeoLatent, we first introduce **Common-Residual Geometry Alignment (CR-GEO)**, which models the geometry shared across teacher features once at the set level and encourages different geometry states to capture complementary residual geometry. This reduces redundant encoding and yields a more differentiated global geometry representation. We further develop **routed optimization**: training first jointly optimizes answer generation and geometric supervision, then uses a temporary visual bottleneck to direct answer-relevant visual information through the latent intermediates, and finally restores full attention while retaining geometric supervision. This allows the model to recover flexible multimodal reasoning while preserving the spatial representations learned during routed training.

Experimental results validate GeoLatent from multiple perspectives. Under identical initialization and batch order, replacing only GeoAnchor’s geometry alignment objective with CR-GEO increases the effective rank of the geometry representation from 1.00 to 3.87. At the bottleneck stage, blocking downstream latent readout reduces accuracy on a fixed set of 128 direction questions from 89.1% to 25.8%, showing that routed optimization makes answer prediction strongly dependent on the latent intermediates. After full visual access is restored, the differentiated geometry representation and the latent-mediated visual route remain available alongside direct image access; removing image content still causes substantial performance degradation, indicating that the final predictions remain strongly dependent on visual evidence. Overall, GeoLatent achieves 73.0% on SPAR-Bench and 72.1% on SPBench, improving over the reported GeoAnchor results by 4.6 and 2.4 points, respectively.

Our main contributions are summarized as follows:

- We analyze decomposed spatial latent reasoning and identify two remaining limitations: GeoAnchor’s geometry alignment can produce highly redundant global geometry repre-

sentations, while latents inserted into the reasoning trajectory can remain underused during answer learning.

- We introduce **GeoLatent**, in which **CR-GEO** separates shared geometry from residual variation to learn more differentiated geometry representations, while **routed optimization** encourages answer-relevant visual information to pass through the latent intermediates during training.
- Representation analyses, intervention experiments, and benchmark evaluations show that GeoLatent learns more differentiated geometry representations, establishes strong dependence on latent intermediates during routed training, and further improves spatial reasoning performance.

2 RELATED WORK

2.1 SPATIAL REASONING IN VISION-LANGUAGE MODELS

Recent work improves spatial reasoning in VLMs mainly through spatial instruction tuning and progressive training (Chen et al., 2024; Cheng et al., 2024; Li et al., 2026d), or by incorporating explicit 2.5D/3D information such as depth, point clouds, and structured spatial representations (Wu et al., 2025a; Zhang et al., 2025a). These approaches provide stronger spatial supervision or geometric cues, but intermediate reasoning is still often expressed in discrete language, which can be insufficient for representing continuous spatial relations. More recently, GeoAnchor (Li et al., 2026c) moves beyond a uniform latent representation by decomposing 3D spatial information into complementary position, direction, and geometry latents with geometric supervision.

2.2 LATENT REASONING AND VISUAL INTERMEDIATES

Latent reasoning replaces verbalized intermediate steps with hidden representations. CODI (Shen et al., 2025) compresses explicit chain-of-thought reasoning into a continuous latent space through self-distillation, while pause-token methods introduce additional internal computation without explicit intermediate symbols (Goyal et al., 2024). In multimodal reasoning, methods such as Mirage and LVR use latent visual intermediates tied to task-specific visual targets, while Aurora introduces predefined perception tokens for visual quantities such as depth and bounding boxes (Yang et al., 2026; Li et al., 2026a; Bigverdi et al., 2025). LIVR (Li et al., 2026e) instead learns latent visual tokens without explicit intermediate supervision and uses visual bottlenecking to force task-relevant visual information through these tokens before answer prediction. RIS (Cui et al., 2026) identifies latent-trajectory collapse and answer bypass and addresses them with spatial-semantic grounding and progressive bottlenecking. Recent mechanistic evidence shows that latent-reasoning gains can reflect boundary markers, formatting, or attention patterns rather than slot content alone (Guo et al., 2026). Accordingly, GeoLatent evaluates both the geometric organization of decomposed spatial latents and their use during answer learning.

2.3 REPRESENTATION COLLAPSE

Representation collapse has been widely studied in self-supervised representation learning (Caron et al., 2021; Zbontar et al., 2021; Bardes et al., 2022). Unlike the instance-agnostic latent-trajectory collapse studied in RIS (Cui et al., 2026), our redundancy arises within an explicitly supervised set of geometry states under coverage-based alignment: balanced utilization does not ensure that different states encode complementary information. GeoLatent addresses this setting with CR-GEO, representing shared geometry at the set level while encouraging individual geometry states to capture residual variation. We use effective rank to quantify the resulting representation diversity (Roy & Vetterli, 2007).

3 METHOD

GeoLatent builds on GeoAnchor’s text-latent interleaved framework with decomposed position (POS), direction (DIR), and geometry (GEO) latents (Li et al., 2026c). On top of this framework, we introduce two key designs: **Common-Residual Geometry Alignment (CR-GEO)** to improve

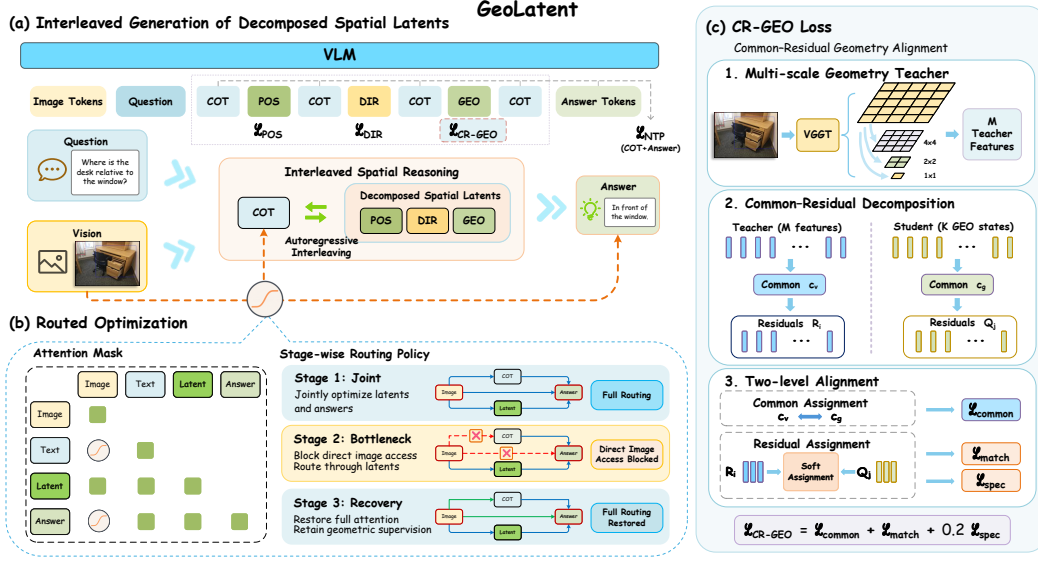


Figure 2: GeoLatent overview. (a) A VLM interleaves language with decomposed POS, DIR, and GEO latents. (b) Routed optimization proceeds through joint, bottleneck, and recovery stages, temporarily restricting direct visual access during answer learning. (c) CR-GEO separates shared from residual teacher geometry for GEO supervision.

the global geometry represented by the GEO latents, and **routed optimization** to promote the use of these latent intermediates during answer learning. Figure 2 illustrates the overall framework.

3.1 PRELIMINARIES: DECOMPOSED SPATIAL LATENTS

GeoLatent follows GeoAnchor’s text–latent interleaved framework (Li et al., 2026c), where language tokens and spatial latents are generated within the same autoregressive reasoning trajectory. Each latent consists of continuous hidden states that are mapped back to the language-model input space by a latent projector, allowing them to condition subsequent generation. Separate geometry heads map the latent hidden states to their corresponding geometric target spaces.

The decomposed latents encode complementary 3D information. POS and DIR represent local cues for object position and relative direction. Their hidden states are mean-pooled and mapped by separate linear heads to a camera-frame 3D position or unit direction vector, supervised with Smooth- ℓ_1 (Girshick, 2015) and cosine losses, respectively. The corresponding targets are precomputed from Depth Anything v3 camera-frame back-projection (Lin et al., 2026). GEO instead represents global scene geometry through multiple latent states projected into the VGTT alignment space and supervised against multi-scale pooled VGTT features (Wang et al., 2025a). Unlike POS and DIR, an individual GEO state is not assigned a predefined physical quantity; supervision is defined over the GEO set and the global teacher geometry.

For training stage s , the overall objective is

$$\mathcal{L}^{(s)} = \mathcal{L}_{\text{NTP}} + \lambda_{\text{local}}^{(s)} (\mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{dir}}) + \lambda_{\text{GEO}}^{(s)} \mathcal{L}_{\text{GEO}}, \quad (1)$$

where $\mathcal{L}_{\text{NTP}}$ supervises the reasoning text and final answer, $\mathcal{L}_{\text{pos}}$ and $\mathcal{L}_{\text{dir}}$ supervise the local spatial latents, and $\mathcal{L}_{\text{GEO}}$ denotes the supervision objective for the global GEO latents. In this paper, we denote GeoAnchor’s original coverage-based GEO objective as $\mathcal{L}_{\text{cov}}$ and our proposed CR-GEO objective as $\mathcal{L}_{\text{CR-GEO}}$. Stage-specific weights and implementation details are given in Appendix B.5. We next examine why the original GEO alignment can still yield highly redundant representations.

3.2 REDUNDANCY UNDER COVERAGE-BASED GEO ALIGNMENT

GeoAnchor aligns multiple GEO states with pooled VGTT teacher features using soft coverage and balanced utilization (Li et al., 2026c). Let $\hat{\mathbf{v}}_i$ denote an ℓ_2 -normalized teacher feature and $\hat{\mathbf{g}}_j$ a

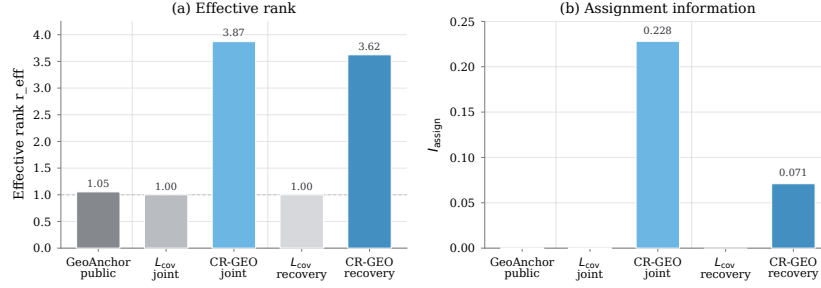


Figure 3: GEO-token organization. The public checkpoint and models trained with the original GEO alignment loss approach rank one with near-zero assignment information, whereas CR-GEO yields differentiated GEO vectors. The joint pair comes from the matched-initialization loss-only comparison; the recovered pair comes from separately trained recovered controls in Table 2.

projected, normalized GEO alignment feature. Define

$$A_{ij} = \frac{\hat{\mathbf{v}}_i^\top \hat{\mathbf{g}}_j}{\tau}, \quad p_{ij} = \text{softmax}_j(A_{i\cdot}), \quad u_j = \frac{1}{M} \sum_i p_{ij}. \quad (2)$$

Ignoring numerical stabilizers, the coverage-based objective is

$$\mathcal{L}_{\text{cov}} = -\frac{1}{M} \sum_i \log \sum_j \exp(A_{ij}) + \lambda_{\text{bal}} \sum_j u_j \log(K u_j), \quad (3)$$

with $\tau = 0.07$ and $\lambda_{\text{bal}} = 0.05$. The first term encourages coverage of the teacher features, while the second encourages balanced marginal use of the K GEO states.

Balanced utilization, however, does not imply differentiated representations. If all GEO alignment features are identical, then $p_{ij} = 1/K$ for all i, j , and consequently $u_j = 1/K$, so the balance term vanishes up to numerical stabilization. Thus, all GEO states may repeatedly encode the same shared geometric component while fully satisfying the utilization constraint.

Proposition 1 (Best collapsed configuration). *Assume $\|\frac{1}{M} \sum_i \hat{\mathbf{v}}_i\|_2 > 0$. Among collapsed configurations $\hat{\mathbf{g}}_1 = \dots = \hat{\mathbf{g}}_K = \mathbf{g}$ with $\|\mathbf{g}\|_2 = 1$, the optimum of Equation 3 is*

$$\mathbf{g}^* = \text{normalize} \left(\frac{1}{M} \sum_i \hat{\mathbf{v}}_i \right), \quad \mathcal{L}_{\text{cov}}(\mathbf{g}^*) = -\log K - \frac{R_V}{\tau}, \quad (4)$$

where $R_V = \|\frac{1}{M} \sum_i \hat{\mathbf{v}}_i\|_2$.

Here, $\hat{\mathbf{g}}_j$ denotes the post-head, ℓ_2 -normalized GEO alignment feature. Appendix A provides the proof. The proposition shows that when the teacher geometry contains a strong shared component, repeatedly aligning all GEO states with that component is explicitly favored within the collapsed family. The objective therefore does not require different GEO states to capture complementary residual geometry, motivating the common-residual formulation below.

3.3 COMMON-RESIDUAL GEOMETRY ALIGNMENT

CR-GEO separates the shared component of teacher geometry from its residual variation. Using the normalized teacher and GEO alignment features above, we define their common directions as

$$\mathbf{c}_v = \text{normalize} \left(\frac{1}{M} \sum_i \hat{\mathbf{v}}_i \right), \quad \mathbf{c}_g = \text{normalize} \left(\frac{1}{K} \sum_j \hat{\mathbf{g}}_j \right). \quad (5)$$

We then remove the teacher common direction from both teacher and student features:

$$\mathbf{R}_i = \hat{\mathbf{v}}_i - (\hat{\mathbf{v}}_i^\top \mathbf{c}_v) \mathbf{c}_v, \quad \mathbf{Q}_j = \hat{\mathbf{g}}_j - (\hat{\mathbf{g}}_j^\top \mathbf{c}_v) \mathbf{c}_v. \quad (6)$$

Let $w_i = \|\mathbf{R}_i\|_2$, and let $\hat{\mathbf{R}}_i$ and $\hat{\mathbf{Q}}_j$ denote the normalized residuals. Residual-to-GEO assignments are

$$a_{ij} = \text{softmax}_j \left(\frac{\hat{\mathbf{R}}_i^\top \hat{\mathbf{Q}}_j}{\tau_r} \right), \quad \tau_r = 0.1. \quad (7)$$

For our microbatch size of one, we write the objective in the following per-example form and omit numerical stabilizers. With $Z = \sum_i w_i$ and $u_j = Z^{-1} \sum_i w_i a_{ij}$, CR-GEO is defined as

$$\begin{aligned} \mathcal{L}_{\text{common}} &= 1 - \mathbf{c}_g^\top \mathbf{c}_v, & \mathcal{L}_{\text{match}} &= 1 - \frac{1}{Z} \sum_{i,j} w_i \text{sg}(a_{ij}) \hat{\mathbf{R}}_i^\top \hat{\mathbf{Q}}_j, \\ \bar{H}_{\text{cond}} &= -\frac{1}{Z \log K} \sum_{i,j} w_i a_{ij} \log a_{ij}, & \bar{B}_{\text{bal}} &= \frac{1}{\log K} \sum_j u_j \log(K u_j), \\ \mathcal{L}_{\text{spec}} &= \bar{H}_{\text{cond}} + 0.1 \bar{B}_{\text{bal}}, & \mathcal{L}_{\text{CR-GEO}} &= \mathcal{L}_{\text{common}} + \mathcal{L}_{\text{match}} + 0.2 \mathcal{L}_{\text{spec}}. \end{aligned} \quad (8)$$

$\mathcal{L}_{\text{common}}$ aligns the shared geometry once at the set level, while $\mathcal{L}_{\text{match}}$ and $\mathcal{L}_{\text{spec}}$ encourage different GEO states to capture complementary residual geometry. The assignments are stop-gradient only in $\mathcal{L}_{\text{match}}$.

3.4 ROUTED OPTIMIZATION

Geometric supervision alone does not ensure that the resulting spatial latents are used during answer learning. Under standard multimodal attention, downstream text and answer tokens can directly access image features and may therefore rely only weakly on the latent intermediates. Inspired by the visual bottlenecking strategy of LIVR (Li et al., 2026e), we introduce a three-stage routed optimization schedule for decomposed spatial latents.

Joint. We first jointly optimize next-token prediction and geometric supervision under standard multimodal attention. This allows the decomposed spatial latents to develop together with reasoning and answer generation.

Bottleneck. We then temporarily prevent question, intermediate-text, and answer queries from attending directly to image keys, while latent queries retain visual access. Visual information needed by subsequent reasoning and answer prediction must therefore pass through the generated spatial latents. Geometric supervision is retained during this stage to preserve their spatial representations.

Recovery. Finally, we restore full visual attention while continuing geometric supervision. The model can again combine direct image access with the latent-mediated route learned during bottleneck training, while retaining the differentiated geometric representations. Detailed routing masks, stage lengths, and loss weights are provided in Appendix B.5.

4 EXPERIMENTS

We evaluate whether the decomposed latents are geometrically structured, used during answer learning, and visually dependent. Objective comparisons and POS/DIR decoding test representation structure; route interventions directly test latent use beyond task accuracy (Guo et al., 2026); benchmark and image interventions evaluate task performance and visual dependence.

4.1 EXPERIMENTAL SETUP

Implementation. GeoLatent uses Qwen3-VL-2B-Instruct (Bai et al., 2025) with two hidden states for each POS/DIR latent and eight for GEO. Joint, bottleneck, and recovery run for 3,311 / 1,000 / 3,311 steps with learning rates $(2, 1, 2) \times 10^{-5}$ and global batch 32. The vision encoder is frozen; the merger, language model, and latent modules are trainable. Controlled runs share the checkpoint, data, optimizer, and phase lengths; the loss-only comparison additionally matches initialization and per-rank batch order.

Training data. We use 105,928 single-image samples: 100k released SPAR records (Zhang et al., 2025b) from ScanNet, ScanNet++, and Structured3D (Dai et al., 2017; Yeshwanth et al., 2023;

Table 1: Spatial-reasoning performance (%). †: results reported by Li et al. (2026c); SPAR numerical entries use threshold-averaged accuracy; others use accuracy. Mean averages the three benchmarks. Best and second-best entries are bold and underlined, respectively.

Model	Params.	SPAR-Bench						SPBench			ViewSpatial	Mean
		Avg	Dep	Dis	Prox	Rel	View	Avg	Rel	Abs	Avg	
<i>Proprietary models reported by Li et al. (2026c)</i>												
GPT-4o [†] (Hurst et al., 2024)	–	40.1	34.3	43.4	57.1	45.9	31.0	53.4	49.4	56.0	37.5	43.7
Gemini-2.5-Flash [†] (Comanici et al., 2025)	–	48.7	37.4	45.2	80.1	63.2	41.0	51.5	44.7	56.0	44.0	48.1
<i>General VLMs reported by Li et al. (2026c)</i>												
MiniCPM-V-4.5 [†] (Yu et al., 2026)	8B	37.7	32.4	32.6	62.1	50.6	29.8	40.9	47.4	36.7	39.0	39.2
LLaVA-OneVision-1.5 [†] (An et al., 2025)	8B	34.5	29.3	32.6	58.5	40.7	26.7	40.8	45.1	38.0	38.7	38.0
Qwen3-VL [†] (Bai et al., 2025)	8B	41.1	32.5	34.6	71.2	62.1	31.5	55.1	47.6	60.0	44.0	46.7
Molmo2 [†] (Clark et al., 2026)	8B	24.6	16.7	3.2	62.4	48.4	25.1	29.7	47.6	18.1	45.5	33.3
GLM-4.1V [†] (Hong et al., 2025)	9B	48.0	42.4	53.2	65.6	57.7	34.3	49.1	43.6	52.6	40.4	45.8
Qwen3.5 [†] (Team, 2026)	9B	48.6	39.0	53.0	73.8	63.5	33.4	48.8	48.6	48.8	<u>45.6</u>	47.7
Kimi-VL [†] (Team et al., 2025)	16B-A3B	33.9	28.6	26.7	60.6	44.5	28.3	42.1	40.8	43.0	38.1	38.0
InternVL3.5 [†] (Wang et al., 2025b)	38B	38.0	31.5	35.5	60.9	58.0	25.5	53.1	49.4	55.5	41.4	44.2
<i>Specialized spatial models</i>												
G2VLM [†] (Hu et al., 2026)	2B	41.7	41.8	66.3	37.7	26.4	24.2	14.3	25.2	7.2	18.3	24.8
3DThinker [†] (Chen et al., 2026)	3B	23.8	15.8	18.6	49.7	24.2	25.1	13.8	32.2	1.7	29.2	22.3
SpatialLadder (Li et al., 2026d)	3B	31.4	20.6	28.1	55.0	41.2	29.8	66.7	73.3	<u>62.5</u>	44.8	47.7
SpatialMLLM-v1.1 [†] (Wu et al., 2025a)	4B	33.7	25.3	34.2	53.8	36.5	31.2	54.2	55.7	53.3	39.7	42.5
ViLaSR [†] (Wu et al., 2025b)	7B	38.6	28.9	43.1	61.8	46.4	28.2	52.2	54.4	50.8	34.3	41.7
Aurora [†] (Bigverdi et al., 2025)	13B	29.6	26.9	29.0	52.7	21.7	25.7	32.1	36.8	29.0	25.6	29.1
<i>Base model and GeoAnchor references</i>												
Qwen3-VL-2B (base) [†]	2B	32.4	21.9	24.0	60.6	49.2	28.3	52.9	51.4	54.8	36.3	40.5
GeoAnchor [†]	2B	<u>68.4</u>	<u>55.5</u>	<u>67.4</u>	<u>80.3</u>	<u>84.6</u>	<u>68.8</u>	<u>69.7</u>	86.7	58.7	47.0	<u>61.7</u>
<i>GeoLatent</i>												
GeoLatent	2B	73.0	58.3	72.3	84.4	87.1	76.3	72.1	<u>84.1</u>	64.3	43.2	62.8

Zheng et al., 2020), plus 5,928 single-image SpatialLadder samples (Li et al., 2026d). POS/DIR targets use Depth Anything v3 camera-frame back-projection; GEO supervision uses VGGT features pooled at {1, 2, 4}.

Evaluation. We evaluate on 2,866 single-image SPAR-Bench examples (Zhang et al., 2025b), 1,009 SPBench examples (Li et al., 2026d), and the 4,607-example single-image subset of ViewSpatial (Li et al., 2026b), following the official prompts and scorers. An overlap audit finds no exact input-image or image/frame-question-answer duplicates under the available metadata; excluding 37 records from two training-overlapping ScanNet scenes changes the three-benchmark mean only from 62.76 to 62.59. Appendix B provides full data construction, training, evaluation, and overlap-audit details.

4.2 MAIN BENCHMARK RESULTS

Task performance and reference-frame transfer. Table 1 shows that GeoLatent reaches 73.0% on SPAR-Bench and 72.1% on SPBench, improving over the reported GeoAnchor results by 4.6 and 2.4 points. All five SPAR categories improve, with the largest gain on View (+7.5 points), while SPBench absolute spatial reasoning improves by 5.6 points. The matched no-GEO control reaches a comparable three-benchmark mean (62.4% versus 62.8%), showing that aggregate accuracy alone cannot determine whether the representation is differentiated or the latent intermediates are used. On ViewSpatial, GeoLatent scores 43.2%, improves camera-relative direction by 5.9 points over the same-data no-GEO control, and achieves the highest Camera score among the matched recovered controls (53.2% versus 48.7–49.4%). Person-centered questions additionally require estimating the reference person’s orientation, which is not explicitly represented by the current latent vocabulary and remains a natural extension. The near-tie in aggregate accuracy therefore motivates separate analyses of representation structure, latent use, and visual dependence.

Table 2: Core objective and routed-optimization controls. Left: GEO objectives at joint and recovered endpoints; right: a 2×2 post-joint comparison of bottleneck training and geometric supervision during recovery. Recovered controls match data, phase lengths, routing, and stage-specific local/GEO supervision weights; the original-objective control therefore uses GeoLatent’s stage-specific weights. Mean is the unweighted average of SPAR-Bench, SPBench, and ViewSpatial.

(a) GEO objective and endpoint					(b) Post-joint route	
Stage	Objective	Mean	I_{assign}	r_{eff}	Route	Mean
Joint	original $\mathcal{L}_{\text{cov}}$	60.0	≈ 0	1.00	Direct NTP	60.0
Joint	CR-GEO	61.3	.228	3.87	Direct recovery	61.2
Recovered	original $\mathcal{L}_{\text{cov}}$	62.3	≈ 0	1.00	Bneck→NTP	60.2
Recovered	CR-GEO	62.8	.071	3.62	Bneck→recovery	62.8

4.3 STRUCTURING THE LEARNED SPATIAL REPRESENTATION

We assess global GEO differentiation using effective rank r_{eff} and assignment information I_{assign} : rank near one and $I_{\text{assign}} \approx 0$ indicate highly redundant states with nearly identical teacher assignments.

Global GEO organization. Figure 3 and Table 2(a) show the core CR-GEO comparison. The matched joint runs start from bit-identical initialization, see the same per-rank batch sequence, and differ only in the GEO objective. The original $\mathcal{L}_{\text{cov}}$ yields $I_{\text{assign}} \approx 0$ and $r_{\text{eff}} = 1.00$, whereas CR-GEO reaches 0.228 and 3.87. This structural change is accompanied by a 1.29-point three-benchmark gain, including a 1.85-point ViewSpatial gain (paired McNemar $p=0.016$). Importantly, the matched objective replacement improves all three benchmarks, so the increased representation diversity is accompanied by task gains rather than occurring only as a diagnostic change. At the separately trained recovered endpoints, the original objective remains rank one, while GeoLatent retains a differentiated representation with $I_{\text{assign}} = 0.071$ and $r_{\text{eff}} = 3.62$, close to the teacher-residual effective rank of approximately 4.1.

Local POS/DIR geometry. The final POS/DIR latents remain quantitatively decodable: on 504 held-out records from 248 training-disjoint scenes, GeoLatent obtains 0.338 m POS error and 0.899 independent-DIR cosine. Replacing only the RGB input with a different-scene image raises POS error from 0.389 to 1.474 m, showing that the decoded local geometry depends on the corresponding image.

Geometry supervision during routed optimization. Table 2(b) forms a 2×2 comparison of bottleneck training and retained geometric supervision. Retaining geometric supervision improves the three-benchmark mean by 1.2 points without bottleneck training and by 2.6 points after bottleneck training, yielding a positive interaction of 1.4 points. The decomposed latent system can therefore retain geometric structure while supporting answer-relevant computation.

4.4 OPERATIONAL USE AND VISUAL DEPENDENCE

Bottleneck-stage latent use. Table 3(a) shows that the bottleneck model retains a 58.0% three-benchmark mean when non-latent answer computation has no direct image access, improving by 3.6 points over the corresponding original- $\mathcal{L}_{\text{cov}}$ reference. On the fixed 128-example SPBench direction set in Figure 4(a), blocking downstream readout from the decomposed latents reduces the matched $K=8$ model from 89.1% to 25.8%, while public GeoAnchor remains at 82.0%. The same intervention changes accuracy by only 0.78 points at the joint checkpoint, by 63.28 points at bottleneck, and again by 0.78 points after recovery. Strong dependence on decomposed latent readout is therefore localized to the bottleneck stage and does not imply the same dependence after recovery.

Recovery route retention. Table 3(b) isolates the image→latent→downstream route after full attention is restored. GeoLatent retains 52.2/51.3/33.5% on SPAR-Bench/SPBench/ViewSpatial under the latent-only route; removing latent readout lowers performance by 5.3/3.7/1.8 points, whereas the matched direct-recovery control shows approximately no corresponding gain. Thus, bottleneck

Table 3: Operational evidence for routed optimization (%). Panel (a) compares the bottleneck endpoints of the main GeoLatent curriculum and the original- $\mathcal{L}_{\text{cov}}$ reference curriculum, where non-latent queries have no direct image access. In panel (b), *Latent-only* removes direct image access for non-latent queries, *Neither* additionally removes latent readout, and direct recovery skips bottleneck training. All recovery-route conditions use the intervention-compatible attention path; its full-attention control therefore differs slightly from the standard decoding path in Table 1.

(a) Bottleneck endpoint comparison

Configuration	SPAR	SPBench	ViewSpatial	Mean
original $\mathcal{L}_{\text{cov}}$ reference	66.4	64.2	32.8	54.5
GeoLatent (CR-GEO)	66.9	67.8	39.3	58.0

(b) Recovered route retention

Model	Route condition	SPAR	SPBench	ViewSpatial
GeoLatent	Full attention	73.24	72.07	43.28
GeoLatent	Latent-only	52.20	51.27	33.49
GeoLatent	Neither	46.89	47.55	31.73
Direct recovery	Latent-only	45.46	43.12	28.91
Direct recovery	Neither	45.52	43.47	28.93

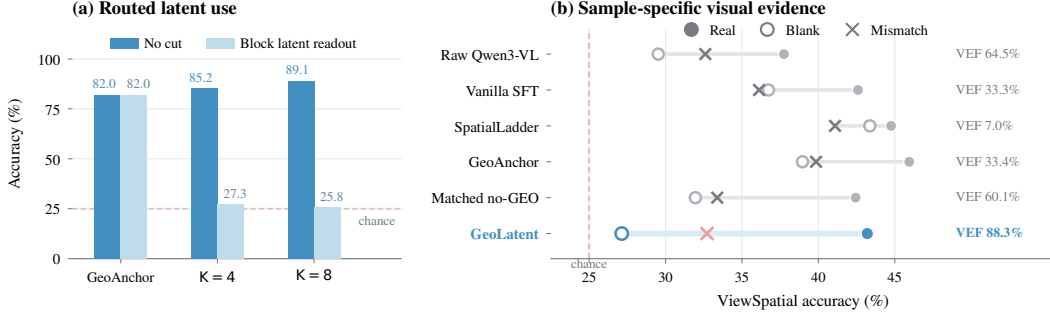


Figure 4: Latent use and sample-specific visual dependence. (a) Blocking downstream readout from all decomposed spatial latents leaves public GeoAnchor unchanged but reduces the matched $K=4/8$ bottleneck models to near-chance accuracy. (b) ViewSpatial under real (filled), blank (open), and mismatched images (cross), including the matched no-GEO control; VEF is the fraction of above-chance accuracy removed by blanking.

training leaves a usable latent-mediated route after recovery, without implying that it remains the dominant path under full attention.

Visual grounding. In Figure 4(b), blanking or mismatching the image reduces GeoLatent’s ViewSpatial accuracy from 43.2% to 27.1% or 32.7%; no-GEO reaches 42.4% with the real image. Blanking removes 88.3% of GeoLatent’s above-chance margin, versus 60.1% for no-GEO. On the Person subgroup, it raises no-GEO from 31.9% to 44.7% but lowers GeoLatent from 28.1% to 24.8%, near chance. Thus, similar aggregate scores mask different grounding: GeoLatent’s useful margin depends more on the corresponding image than on the Person-template prior. Substantial dependence on the corresponding image is also observed on SPAR-Bench and SPBench.

More comprehensive analyses. Due to space constraints, full ablations, cross-backbone and GEO-capacity studies, latent diagnostics, route-localization controls, visual-prior analyses, and statistical details are provided in Appendix B.

5 CONCLUSION

GeoLatent addresses redundant GEO representations and underuse of decomposed spatial latents. CR-GEO separates shared from residual teacher geometry, while routed optimization temporarily directs visual answer learning through the latents before restoring full attention. Experiments show differentiated GEO representations, decodable image-specific POS/DIR latents, strong bottleneck-stage latent use, and a usable latent-mediated route after recovery. Image interventions further show that the recovered predictions remain tied to corresponding visual evidence rather than the ViewSpatial Person-template prior. Together, these results improve spatial reasoning while separating representation structure, latent use, and visual dependence.

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TECHNICAL APPENDICES AND SUPPLEMENTARY MATERIAL

A PROOF OF PROPOSITION 1

For any collapsed normalized alignment configuration $\hat{\mathbf{g}}_1 = \dots = \hat{\mathbf{g}}_K = \mathbf{g}$, the balance term vanishes and the remaining objective is $-\log K - \mathbf{g} \cdot (\frac{1}{M} \sum_i \hat{\mathbf{v}}_i) / \tau$. Define

$$\bar{\mathbf{v}} = \text{normalize} \left(\frac{1}{M} \sum_i \hat{\mathbf{v}}_i \right).$$

By Cauchy–Schwarz, the objective is minimized by $\mathbf{g} = \bar{\mathbf{v}}$. With $\hat{\mathbf{g}}_j^* = \bar{\mathbf{v}}$ for all j , all cosine similarities are equal: $A_{ij} = \hat{\mathbf{v}}_i \cdot \bar{\mathbf{v}} / \tau$ for every j . Hence $\sum_j \exp(A_{ij}) = K \exp(\hat{\mathbf{v}}_i \cdot \bar{\mathbf{v}} / \tau)$ and

$$-\frac{1}{M} \sum_i \log \sum_j \exp(A_{ij}) = -\log K - \frac{1}{M\tau} \sum_i \hat{\mathbf{v}}_i \cdot \bar{\mathbf{v}} = -\log K - \frac{1}{\tau} \left\| \frac{1}{M} \sum_i \hat{\mathbf{v}}_i \right\|,$$

where the last step uses the definition of $\bar{\mathbf{v}}$. The assignment of every teacher feature is uniform over the K latents, so $u_j = 1/K$ and the balance term vanishes. $\square$

Numerically, on the frozen evaluation frame used for the representation diagnostics, the absolute difference between the analytical and computed collapsed-configuration losses is 1.4×10^{-6} on average. The collapsed configuration attains a lower loss than the learned latents on 100% of samples for every checkpoint tested (collapsed minus learned: public release -0.82 , original- $\mathcal{L}_{\text{cov}}$ joint with reference weights -0.78 , and original- $\mathcal{L}_{\text{cov}}$ recovery with reference weights -0.05). The teacher-feature spherical k -means baseline is worse on 100% of samples (teacher-feature spherical k -means minus collapsed: $+0.198$).

B EXPERIMENTAL DETAILS AND ADDITIONAL RESULTS

B.1 DATA AND BENCHMARK CONSTRUCTION

All training and evaluation samples are single-image examples. Local POS and DIR targets use the camera coordinate frame: Depth Anything v3 (DA3) depth is back-projected with camera intrinsics and robust median aggregation inside each marked region, and DIR targets are normalized differences between the corresponding positions. The same construction is applied to every source.

Spatial reasoning training mixture. Table 4 summarizes the 105,928-sample training mixture used by the main GeoLatent runs. It follows the source composition of the GeoAnchor spatial reasoning data: SPAR-derived ScanNet, ScanNet++, and Structured3D samples plus SpatialLadder single-image samples. In the released SPAR records, Structured3D spatial-imagination questions use sentence-format answers; we retain that released format for both corresponding buckets.

Geometry supervision. In geometry-enabled configurations, records may contain local POS/DIR spans and a global GEO span. POS and DIR receive local geometric supervision when the corresponding objects are present, while the GEO span is aligned to VGGT teacher features. Following GeoAnchor, we use the final VGGT feature map and apply average pooling at resolutions $\{1, 2, 4\}$, giving $1^2 + 2^2 + 4^2 = 21$ teacher vectors per image. GeoAnchor’s GEO loss aligns the eight vectors within the GEO token to these teacher features with soft coverage and balance; GeoLatent replaces this alignment term with CR-GEO while leaving the teacher features and pooled resolution unchanged.

B.2 BENCHMARK DEFINITIONS

SPAR-Bench and SPBench are in-domain with respect to the spatial reasoning sources, while ViewSpatial serves as an external cross-dataset transfer benchmark with distinct examples and partially overlapping spatial task definitions. Table 5 lists the task groups and sample counts used by the evaluation harness.

Table 4: Spatial reasoning training mixture. “Select” denotes multiple-choice answers and “numeric” denotes fill-in numerical answers. Structured3D spatial-imagination is sentence-format in the released SPAR records.

Source	Task group	Answer format	Count
ScanNet / SPAR	depth_prediction.oc	numeric	4,000
	depth_prediction.oo	numeric	4,000
	distance_prediction.oc	numeric	4,000
	distance_prediction.oo	numeric	4,000
	distance_infer_center.oo	select	8,000
	obj_spatial_relation.oo	select	8,000
	spatial_imagination.oc	select	4,000
	spatial_imagination.oo	select	4,000
ScanNet++ / SPAR	depth_prediction.oc	numeric	4,000
	depth_prediction.oo	numeric	4,000
	distance_prediction.oc	numeric	4,000
	distance_prediction.oo	numeric	4,000
	distance_infer_center.oo	select	4,000
	obj_spatial_relation.oo	select	4,000
	spatial_imagination.oc	select	4,000
	spatial_imagination.oo	select	4,000
Structured3D / SPAR	depth_prediction.oc	numeric	4,000
	depth_prediction.oo	numeric	4,000
	distance_prediction.oc	numeric	4,000
	distance_prediction.oo	numeric	8,000
	spatial_imagination.oc	sentence	4,000
	spatial_imagination.oo	sentence	4,000
SpatialLadder-26K	relative direction	select	2,253
	absolute distance	numeric	1,127
	object size	numeric	1,514
	relative distance	select	1,034
Total	–	–	105,928

Table 5: Benchmark task groups and sample counts. SPAR-Bench follows the single-image subset used in GeoAnchor. ViewSpatial additionally tests transfer from camera-frame supervision to camera- and person-centered reference frames.

Benchmark	Reported group	Underlying task(s)	Count
SPAR-Bench	Dep.	depth_prediction.oc, depth_prediction.oo	732
	Dis.	distance_prediction.oc, distance_prediction.oo	756
	Prox.	distance_infer_center.oo	340
	Rel.	obj_spatial_relation	364
	View	spatial_imagination.oc, spatial_imagination.oo	674
SPBench	Rel.	object_rel.direction, object_rel.distance	397
	Abs.	object_abs.distance, object_size.estimation	612
ViewSpatial	Camera–relative direction	camera-view relative direction	1,773
	Camera–object view	camera-view object orientation	996
	Person–object view	person-centered object/person orientation	996
	Person–relative direction	person-centered relative direction	842
Total	SPAR-Bench / SPBench / ViewSpatial	–	2,866 / 1,009 / 4,607

For SPAR-Bench, multiple-choice questions are scored by exact option correctness, while numerical questions use the benchmark’s threshold-averaged accuracy over distance/depth tolerances. SP-Bench reports relative and absolute spatial categories. ViewSpatial reports camera-perspective and person-perspective subgroups; the main paper uses its overall average, while Appendix analyses use the four subgroups to characterize how the learned camera-frame geometry transfers across reference frames.

Train–benchmark overlap audit. We compare the final 105,928-record training mixture against every evaluated example using record IDs, normalized questions, exact input-image SHA-256, and, where available, ScanNet scene/frame identity. Table 6 separates shared domains and templates from duplicated examples. No benchmark record shares an exact input image or an image/frame–question–answer composite with training. SPAR-Bench does not expose scene/frame metadata, so no scene-disjoint claim is made for that benchmark. Its exact image and composite checks re-

main zero. SPBench and ViewSpatial contain 23 and 14 records from two ScanNet scenes also present in the SPAR-derived training source; removing those records changes SPBench from 72.11 to 71.74%, ViewSpatial from 43.20 to 43.04%, and the GeoLatent three-benchmark mean from 62.76 to 62.59%. Repeated question strings in SPAR-Bench and SPBench arise on different images and reflect their templated, in-domain task definitions rather than duplicated QA examples.

Table 6: Overlap of each evaluation set with the 105,928-record training mixture. Entries count affected benchmark records. “Exact example” requires matching image or scene/frame, normalized question, and answer. A dash means the benchmark does not expose that metadata.

Benchmark	Record ID	Scene	Scene/frame	Image SHA	Question	Exact example
SPAR-Bench	0	—	—	0	63	0
SPBench	0	23	5	0	777	0
ViewSpatial	0	14	2	0	0	0

B.3 FULL BLANK- AND MISMATCHED-IMAGE INTERVENTIONS

For the blank condition, we replace every benchmark image with a same-size mid-gray canvas. For the mismatched condition, each image is replaced according to one fixed donor mapping with a different natural image from the same benchmark and task/answer-format group. Donors always have a different image hash; they additionally come from a different scene whenever scene metadata is available. Questions, choices, decoding configuration, and scorer are unchanged within each comparison. Trained models require a complete `<answer>` span, whereas the raw instruct model uses the official flexible parser because it was not trained with this tagged protocol. “Vanilla SFT” is a same-data, answer-only reference without decomposed spatial latents, adapted using LoRA (Hu et al., 2022) ($r=32$, $\alpha=64$, learning rate 2×10^{-5} , two epochs). All runs use the complete fixed denominator.

Table 7: Visual-evidence interventions. Panel (a) reports full-benchmark accuracy/score as real/blank/mismatched. Panel (b) decomposes ViewSpatial into Camera and Person aggregates. Direct recovery skips the bottleneck; SpatialLadder uses matched pixel bounds.

(a) Full benchmark matrix (real / blank / mismatched)			
Model	SPAR-Bench	SPBench	ViewSpatial
Raw Qwen3-VL	28.97/20.83/26.36	52.06/23.48/34.72	37.75/29.52/32.60
Vanilla SFT	62.01/41.94/39.44	62.38/41.85/41.86	42.59/36.73/36.12
SpatialLadder	31.44/30.80/29.10	66.75/45.95/43.86	44.76/43.37/41.09
GeoAnchor	67.66/44.23/39.13	69.22/43.52/41.62	45.95/38.96/39.85
Matched no-GEO	72.92/47.45/38.34	71.78/41.57/42.53	42.44/31.95/33.38
Original L_{cov}	73.21/45.90/38.21	70.43/41.62/43.71	43.22/27.41/34.17
Direct recovery	72.08/45.62/38.20	71.15/41.17/43.12	40.42/26.89/30.61
GeoLatent	72.98/46.03/38.07	72.11/43.82/44.53	43.20/27.13/32.71

(b) ViewSpatial reference-frame aggregates						
Model	Camera			Person		
	Real	Blank	Mismatch	Real	Blank	Mismatch
Raw Qwen3-VL	38.82	34.34	34.20	36.13	22.25	30.20
Vanilla SFT	37.92	30.01	27.16	49.62	46.84	49.62
SpatialLadder	38.35	31.53	29.97	54.41	61.21	57.83
GeoAnchor	45.14	31.17	33.80	47.17	50.71	48.97
Matched no-GEO	49.44	23.47	31.78	31.88	44.72	35.80
Original L_{cov}	48.72	22.32	31.02	34.93	35.09	38.90
Direct recovery	49.19	29.40	31.64	27.20	23.12	29.05
GeoLatent	53.20	28.67	34.16	28.13	24.81	30.52

Visual dependence is not confined to ViewSpatial. Matched no-GEO loses 25.5/30.2 points under blanking and 34.6/29.3 under mismatched images on SPAR-Bench/SPBench. The control using GeoAnchor’s GEO loss and the direct-recovery control fall from 62.28/61.22% real-image three-benchmark means to 38.31/37.89% when blanked and 38.69/37.31% when mismatched. GeoLatent loses 27.0/28.3 points under blanking on SPAR-Bench/SPBench and 34.9/27.6/10.5 under mismatched images on SPAR-Bench/SPBench/ViewSpatial; for each benchmark, no sampled sign flip is as extreme as observed (100,000 draws; plus-one $p=1/100001$). Thus the closely scoring

recovered controls remain strongly dependent on visual input; GeoLatent combines this evidence dependence with a differentiated GEO representation and the highest ViewSpatial Camera score among the evaluated models and controls.

SpatialLadder exhibits a different benchmark-dependent pattern: real/blank/mismatched changes only from 31.44/30.80/29.10% on SPAR-Bench, but from 66.75/45.95/43.86% on SPBench. On ViewSpatial it loses only 3.7 points under mismatched images overall (44.76 $\rightarrow$ 41.09), with the loss concentrated in Camera questions while Person accuracy remains above its real-image score.

The reference-frame decomposition localizes ViewSpatial’s language prior. Vanilla SFT loses only 2.78 points on Person questions, while SpatialLadder, GeoAnchor, and matched no-GEO improve after blanking; no-GEO separates sharply into Camera 49.44 $\rightarrow$ 23.47% and Person 31.88 $\rightarrow$ 44.72%. Thus its strong in-domain accuracy is visually supported, but part of its ViewSpatial Person score remains available without the corresponding image. SpatialLadder’s Camera questions show the opposite pattern: real 38.35 versus 31.53 blank and 29.97 mismatched. In contrast, GeoLatent’s blank Person score is 24.81%, effectively the four-way chance level. Its real-image strength is concentrated in Camera questions, where it reaches 53.20%—8.1 points above GeoAnchor—and falls by 24.53 points when the image is removed. Thus GeoLatent’s ViewSpatial predictions derive their useful margin from the visually grounded reference frame rather than the Person-template prior. The following audit connects this behavior to the benchmark’s question and answer distributions.

B.4 TEMPLATE AND ANSWER PRIORS IN VIEWSPATIAL

We audit the 20,680 spatial QA records in SpatialLadder-26K against the 4,607 ViewSpatial questions used in our single-image evaluation. SpatialLadder contains 2,253 single-image camera-relative-direction questions of the form “From the camera’s perspective, is object A to object B’s left, right, front, or back?” ViewSpatial uses surface paraphrases of the same relation-and-reference-frame construct for its 1,773 Camera-Relative-Direction questions. SpatialLadder also contains 2,162 multi-view questions of the form “standing by A and facing B, where is C?”, closely matching ViewSpatial’s scene-simulation family. The latter family is outside our 4,607-question four-task subset, but demonstrates that the two resources share a common template vocabulary and reference-frame construction.

After lowercase and punctuation normalization, no complete question string is shared between the 20,680 SpatialLadder records and the 4,607 ViewSpatial questions. The exploitable signal is therefore a task-distribution prior rather than verbatim QA duplication. Table 8 separates answer-position balance from answer-semantics balance.

Table 8: ViewSpatial answer priors. “Letter max” is the frequency of the most common correct option position; “semantic max” is the most frequent direction irrespective of option position.

Task	N	Letter max	Semantic max	Direction
Camera-relative direction	1,773	26.4	16.0	right
Camera-object orientation	996	26.0	26.8	back
Person-object orientation	996	26.1	57.2	front
Person-relative direction	842	34.8	27.4	right

The strongest prior occurs in Person-Object-Orientation: 570 of 996 gold answers are semantically “front,” although their A-D positions remain balanced. With blank images, SpatialLadder predicts “front” on 614 examples, “front-left” on 241, and “front-right” on 81, reaching 76.10% on this subgroup and 74.20% with mismatched images. This behavior is consistent with reliance on direction semantics and template-conditioned option structure rather than a fixed answer letter. Its overall accuracy is 43.37% with blank images and 41.09% with mismatched images, compared with 44.76% on real images.

GeoLatent provides an informative contrast because its training mixture includes all 5,928 SpatialLadder single-image examples, including the 2,253 camera-relative questions. Nevertheless, blanking ViewSpatial images reduces GeoLatent from 43.20% to 27.13%, close to the 25% chance level. Exposure to the template family therefore does not by itself account for GeoLatent’s normal-image performance: its recovered predictions derive most of their above-chance ViewSpatial accuracy from visual evidence.

B.5 TRAINING CONFIGURATION

Table 9 summarizes the three training phases and their stage-specific objective weights.

Table 9: GeoLatent training phases. All phases use Qwen3-VL-2B-Instruct, frozen vision encoder, trainable merger/LLM/latent modules, global batch 32, bf16 computation, and fp32 optimizer states.

Phase	Steps	LR	Attention policy	λ_{NTP}	λ_{local}	λ_{GEO}
Joint	3,311	2×10^{-5}	full	1.0	1.0	0.2
Bottleneck	1,000	1×10^{-5}	decomposed-latent bottleneck	1.0	0.1	0.1
Recovery	3,311	2×10^{-5}	full	1.0	0.1	0.2

Reference configuration with original $\mathcal{L}_{\text{cov}}$. The reference configuration shares the data, Qwen3-VL-2B-Instruct initialization, phase lengths, and learning rates, with local/GEO supervision weights 1.0/0.1 in joint training, 0.1/0.01 in bottleneck training, and 0.01/0.01 in recovery. GeoLatent uses 1.0/0.2, 0.1/0.1, and 0.1/0.2, respectively. These rows compare full training curricula; the matched-initialization experiment below isolates the GEO loss itself.

Matched-initialization objective test. For the loss-only joint test, we construct one shared seed-3407 initialization after all added latent modules are instantiated, then launch both arms from those bit-identical tensors. The per-rank batch sequences, data, optimizer, learning rate, step count, attention policy, and stage-specific local/GEO supervision weights are identical; the only changed field is $\mathcal{L}_{\text{CR-GEO}}$ versus $\mathcal{L}_{\text{cov}}$. The shared-init fingerprint and the first 64 batches on every rank match exactly. These are the loss-only joint rows of Table 2. The recovered endpoints match data, phase lengths, routing schedule, and GeoLatent’s stage-specific local/GEO supervision weights, but are trained separately under different GEO objectives; they are not continuations of the shared-initialization joint pair.

During bottleneck training, question, intermediate-text, and answer queries cannot directly attend to image keys, while the decomposed latent queries retain image access. Recovery restores full attention while continuing the corresponding local and GEO supervision. These controls isolate how routed answer learning and retained geometric supervision affect the learned latent representations.

B.6 FULL OBJECTIVE AND CURRICULUM ABLATION

All rows in Table 10 use the same 105,928 samples, Qwen3-VL-2B-Instruct base checkpoint, and initialization procedure unless otherwise noted.

Table 3(a) reports the bottleneck endpoints in the main text. GeoLatent preserves strong in-domain spatial reasoning while operating through the decomposed latents and improves the three-benchmark mean by approximately 3.6 points over the corresponding model using GeoAnchor’s GEO loss. This end-to-end stage comparison complements the matched-initialization joint test, which isolates the GEO loss itself.

Local-only recovery and recovery with local/GEO weights 0.01/0.05 improve over bottleneck-to-NTP recovery by 1.5–1.7 points, while the standard GeoLatent recovery weights 0.1/0.2 yield the largest gain. The full-curriculum controls further show that recovery can restore task performance under the original objective, whereas in this comparison only the CR-GEO configuration builds and retains a differentiated GEO representation.

The matched-initialization joint rows isolate the GEO-loss effect. For these fixed checkpoints, CR-GEO improves SPAR-Bench, SPBench, and ViewSpatial by 0.9, 1.1, and 1.85 points, respectively, for a 1.29-point gain in their equal-weight mean; the paired ViewSpatial difference is +1.85 points (McNemar $p=0.016$, bootstrap 95% CI [+0.35, +3.34]). The same starting tensors diverge to $(I_{\text{assign}}, r_{\text{eff}}) = (0.228, 3.87)$ under CR-GEO and approximately $(0, 1.00)$ under GeoAnchor’s GEO loss. The separately trained full-curriculum endpoints show the same structural distinction: the original-objective control using GeoLatent weights remains rank one, while GeoLatent reaches $(0.071, 3.62)$ after bottleneck and recovery.

Table 10: Full loss and curriculum matrix. The post-joint recovery variants use local/GEO supervision weights 0/0 (NTP-only), 0.1/0 (local-only), 0.01/0.05, and 0.1/0.2 (standard GeoLatent recovery). “Reference weights” denotes the stage-specific weights of the original- $\mathcal{L}_{\text{cov}}$ reference configuration, while “GeoLatent weights” denotes GeoLatent’s stage-specific local/GEO supervision weights.

GEO objective	Training route	SPAR	SPBench	ViewSpatial	Mean
<i>Matched-initialization loss-only joint test</i>					
original GEO $\mathcal{L}_{\text{cov}}$	joint only	68.6	70.1	41.3	60.0
CR-GEO $\mathcal{L}_{\text{CR-GEO}}$	joint only	69.6	71.2	43.1	61.3
<i>Additional independently trained joint checkpoints</i>					
no GEO span/objective	joint only	68.4	69.4	43.0	60.3
original GEO $\mathcal{L}_{\text{cov}}$ (reference weights)	joint only	68.3	69.9	40.5	59.6
original GEO $\mathcal{L}_{\text{cov}}$ (GeoLatent weights; independent init)	joint only	67.9	63.6	36.9	56.1
CR-GEO $\mathcal{L}_{\text{CR-GEO}}$ (independent init)	joint only	68.1	70.4	44.2	60.9
<i>Post-joint controls</i>					
no GEO span/objective	bottleneck $\rightarrow$ recovery	72.9	71.8	42.4	62.4
original GEO $\mathcal{L}_{\text{cov}}$ (reference weights)	bottleneck $\rightarrow$ recovery	73.3	71.3	42.2	62.2
original GEO $\mathcal{L}_{\text{cov}}$ (GeoLatent weights)	bottleneck $\rightarrow$ recovery	73.2	70.4	43.2	62.3
original GEO $\mathcal{L}_{\text{cov}}$	direct NTP continuation	72.4	72.0	44.2	62.9
CR-GEO	direct NTP continuation	72.8	70.7	36.6	60.0
CR-GEO	direct recovery (no bottleneck)	72.1	71.2	40.4	61.2
CR-GEO	bottleneck $\rightarrow$ NTP-only recovery	71.6	70.8	38.2	60.2
CR-GEO	bottleneck $\rightarrow$ local-only recovery	72.4	71.5	41.8	61.9
CR-GEO	bottleneck $\rightarrow$ recovery (0.01/0.05)	73.3	71.7	40.3	61.8
CR-GEO	bottleneck $\rightarrow$ recovery	73.0	72.1	43.2	62.8

GEO-span capacity. Table 11 varies only the number of vectors in the GEO token. The four runs share the same backbone initialization, data and batch order, random seed, objective weights, routing schedule, and optimization hyperparameters; K -specific parameters follow the same initialization rule. The frozen K -aware diagnostic frame has teacher-residual effective rank 3.76.

Table 11: GEO-vector count ablation under matched full-curriculum runs. Stage columns report the three-benchmark mean; recovery columns give individual benchmark scores. Throughput is measured during recovery.

K	Joint	Bneck	SPAR	SPB	VS	Rec. mean	r_{eff}	Samples/s
2	61.2	56.5	72.8	71.9	43.9	62.9	1.70	3.37
4	61.3	53.7	74.1	71.9	45.0	63.6	3.23	2.86
8	60.4	55.9	72.8	72.0	42.5	62.4	3.81	2.46
16	60.7	52.9	73.2	72.2	42.1	62.5	4.05	1.77

All four settings remain non-collapsed. $K=2$ is task-effective but dimensionally constrained ($r_{\text{eff}}=1.70$); $K=4$ approaches the teacher’s residual capacity and gives the best recovery mean. Larger spans saturate the available structure without improving task performance, while recovery throughput falls from 2.86 samples/s at $K=4$ to 1.77 at $K=16$. This sweep suggests $K=4$ as a favorable compact operating point rather than a monotonic benefit from adding vectors. The main $K=8$ run was fixed for compatibility with the inherited GeoAnchor interface and belongs to a separate controlled run set, so the two $K=8$ endpoints are not directly comparable.

B.7 CROSS-BACKBONE REPRODUCTION

We reproduce the joint–bottleneck–recovery pipeline with Qwen2.5-VL-3B. The two arms start from the same initialization, use the same 105,928 examples in the same batch order, and share all optimization and routing settings; only the GEO alignment loss differs. Both recovered endpoints use a learning rate of 10^{-5} . Table 12 shows that the structural distinction is reproduced on a second backbone: the model trained with GeoAnchor’s GEO loss has a rank-one GEO span, whereas CR-GEO retains nearly four effective dimensions and nonzero teacher–GEO-vector assignment information. CR-GEO also improves both in-domain benchmarks and the three-benchmark mean, while ViewSpatial decreases by 1.70 points.

Local POS/DIR decodability also transfers: CR-GEO obtains 0.396 m POS error and 0.888 DIR cosine on the frozen held-out set. Replacing RGB with a different-scene image raises POS error

Table 12: Matched cross-backbone reproduction with Qwen2.5-VL-3B. Benchmark entries are accuracy/score (%); structure is measured on the same frozen 768-frame protocol used for the main backbone. Mean is the unweighted average of the three benchmark scores.

GEO objective	SPAR	SPBench	ViewSpatial	Mean	r_{eff}	I_{assign}
original $\mathcal{L}_{\text{cov}}$	67.90	66.14	40.53	58.19	1.00	≈ 0
CR-GEO $\mathcal{L}_{\text{CR-GEO}}$	68.63	67.83	38.83	58.43	3.84	0.178

from 0.444 to 1.477 m and lowers DIR cosine from 0.922 to 0.219, confirming that the decoded local tokens depend on the corresponding input image.

B.8 ADDITIONAL DIAGNOSTICS AND QUALITATIVE VISUALIZATIONS

Figure 5 provides complementary qualitative views of latent spread and local-token attention.

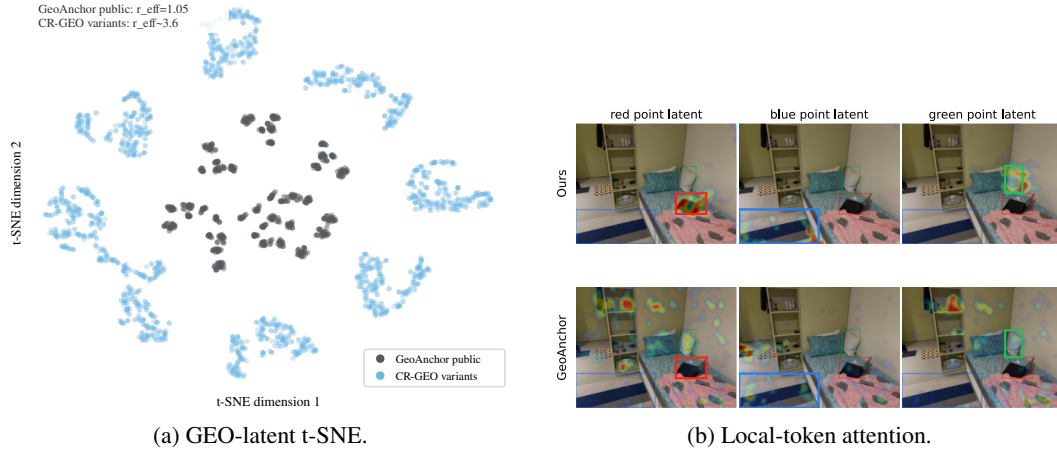


Figure 5: Qualitative structure of the learned spatial latents. The t-SNE projection illustrates reduced concentration of GEO vectors under CR-GEO, and the local-token maps show object-specific visual attention during latent generation. Quantitative structure and use are reported in Table 14 and Figure 4.

B.8.1 LOCAL-TOKEN DECODING ACROSS THE CURRICULUM

We evaluate the six matched objective-stage checkpoints on 504 records from 248 scenes with zero scene, image, or record-ID overlap with the 105,928 training examples. The frame contains 1,134 POS and 79 DIR targets in the camera coordinate system. Table 13 reports teacher-forced and live-generation measurements under each checkpoint’s native routing mode.

Both objectives exhibit a temporary bottleneck-stage increase in POS error followed by full recovery. For CR-GEO, recovery changes POS by -0.011 m relative to joint training (scene-bootstrap 95% CI $[-0.020, -0.0001]$) and changes DIR cosine by -0.024 (CI $[-0.068, 0.001]$). Across the six checkpoints, 176–185 of the 192 records yield aligned live measurements; among these successfully aligned cases, latent-span closure is 100%. The final CR-GEO model also substantially exceeds train-derived constant priors (POS L2 1.066 m; DIR cosine 0.113). The POS prior is the mean of all training POS targets, and the DIR prior is the mean of all training DIR targets normalized for cosine evaluation; held-out evaluation data are not used to construct either prior. These comparisons establish POS/DIR decodability beyond a category or dataset mean.

For normalized teacher features $\hat{\mathbf{v}}_i$ and GEO vectors $\hat{\mathbf{g}}_j$, define the soft assignment $p_{ij} = \text{softmax}_j(\hat{\mathbf{v}}_i^\top \hat{\mathbf{g}}_j / \tau)$ with $\tau = 0.07$ and its marginal $u_j = M^{-1} \sum_i p_{ij}$. We report

$$I_{\text{assign}} = \frac{H(\mathbf{u}) - M^{-1} \sum_i H(\mathbf{p}_i)}{\log K}, \quad (9)$$

Table 13: POS/DIR decoding across the matched 2×3 objective-stage grid. TF denotes teacher-forced evaluation. POS is mean per-point L2 error in meters; DIR and composed DIR are cosine similarities to ground truth. Live/replay DIR agreement is the cosine agreement between the DIR representation captured during live generation and the DIR representation obtained by teacher-forced replay of the same generated sequence. Live evaluation starts from the fixed 192-record subset; reported live metrics use the successfully aligned cases.

Objective	Stage	TF POS ↓	TF DIR ↑	Composed DIR ↑	Live POS ↓	Live DIR ↑	Live/replay DIR agreement ↑
CR-GEO	joint	0.347	0.923	0.871	0.307	0.942	1.000
CR-GEO	bottleneck	0.365	0.911	0.864	0.345	0.919	0.971
CR-GEO	recovery	0.338	0.899	0.874	0.300	0.920	1.000
original $\mathcal{L}_{\text{cov}}$	joint	0.354	0.890	0.866	0.310	0.909	1.000
original $\mathcal{L}_{\text{cov}}$	bottleneck	0.376	0.876	0.862	0.345	0.884	0.957
original $\mathcal{L}_{\text{cov}}$	recovery	0.335	0.907	0.868	0.305	0.916	1.000

which is zero when every teacher feature induces the same assignment distribution and increases when assignments are vector-specific and collectively distributed across the span. It is an assignment diagnostic, not a general neural mutual-information estimator.

We additionally report $H_{\text{cond}} / \log K$, the normalized teacher-conditioned assignment entropy, and

$$N_{\text{LSE}} = \text{mean}_i \exp \left(\log \sum_j e^{A_{ij}} - \max_j A_{ij} \right), \quad (10)$$

which ranges approximately from 1 to K and measures the effective number of covering vectors.

Table 14: GEO-token representation diagnostics on frozen samples ($n=768$). r_{eff} is the effective rank of the 8×8 cosine Gram matrix. I_{assign} is normalized information induced by teacher-to-vector assignments, not a general neural mutual-information estimator. N_{LSE} is the effective number of covering vectors. Dashes denote diagnostics unavailable from historical captures, not failed measurements.

Model	off-diag cos	r_{eff}	$H_{\text{cond}} / \log K$	I_{assign}	N_{LSE}
original $\mathcal{L}_{\text{cov}}$, matched-init joint	1.000	1.00	—	≈ 0	—
CR-GEO, matched-init joint	0.167	3.87	—	0.228	—
GeoAnchor public	0.991	1.05	0.999	≈ 0	7.54
original $\mathcal{L}_{\text{cov}}$, joint (reference weights)	0.460	2.73	0.667	≈ 0	4.00
original $\mathcal{L}_{\text{cov}}$, recovery (reference weights)	1.000	1.00	1.000	≈ 0	7.99
original $\mathcal{L}_{\text{cov}}$, joint (GeoLatent weights)	1.000	1.00	—	≈ 0	—
original $\mathcal{L}_{\text{cov}}$, recovery (GeoLatent weights)	1.000	1.00	—	≈ 0	—
CR-GEO, main-curriculum joint	—	2.72	—	0.048	—
CR-GEO, direct NTP continuation	0.425	2.85	0.803	0.043	3.69
CR-GEO, bottleneck $\rightarrow$ NTP-only	0.216	3.55	0.649	0.055	2.43
GeoLatent, recovery	0.224	3.62	0.579	0.071	2.00

B.8.2 DECOMPOSED-LATENT ROUTE INTERVENTIONS

We apply attention-edge interventions to independently trained bottleneck models with four or eight GEO vectors on the same frozen 64 scenes (128 directions). “Block P/D” prevents subsequent non-latent queries from reading POS and DIR key/value slots; “Block latent readout” additionally blocks downstream non-latent readout of GEO keys while preserving latent generation and latent-to-latent relay. Token identities, projector outputs, images, and the prefix preceding each intervention are unchanged. All conditions use a fixed denominator, and mask-difference checks verify that only the intended edges change.

On the same frozen pairs and under the same four-dimensional additive attention-mask protocol (batch, head, query, key), the public GeoAnchor checkpoint remains at 82.03% before and after the latent-readout cut. Replication across K supports the stage-level mechanism on this frozen direction subset: once the direct visual path is closed, answers depend on the decomposed latent

Table 15: Decomposed-latent localization under bottleneck inference. Entries are accuracy (drop from the no-cut condition), in percent. “Block prior-latent access” prevents all subsequent typed and non-typed queries from attending to previously generated typed-latent keys. For both $K=4$ and $K=8$, neither the POS+DIR drop nor the latent-readout drop was matched or exceeded by any of 20,000 two-sided scene-clustered permutations (plus-one $p=1/20001$ for each).

Condition	$K=4$		$K=8$	
	Acc.	Drop	Acc.	Drop
No latent-readout cut	85.16	–	89.06	–
Block POS	61.72	23.44	53.12	35.94
Block DIR	81.25	3.91	85.94	3.13
Block POS+DIR	26.56	58.59	26.56	62.50
Block GEO	85.16	0.00	89.06	0.00
Block latent readout	27.34	57.81	25.78	63.28
Block prior-latent access	22.66	62.50	27.34	61.72

intermediates. On this direction subset, jointly cutting POS and DIR removes nearly all above-chance accuracy. This dependence on the latent intermediates is the targeted effect of bottleneck training.

Table 16: Stage-wise route intervention in the matched $K=8$ trajectory on the same 128 directions. “Block latent readout” hides all decomposed-latent keys from subsequent non-latent queries while preserving latent-to-latent relay.

Checkpoint	No latent-readout cut	Block latent readout	Drop
Joint (full attention)	69.53	68.75	0.78
Bottleneck (routed)	89.06	25.78	63.28
Recovery (full attention)	82.03	81.25	0.78

The intervention changes accuracy by only 0.78 points at joint, by 63.28 points at bottleneck, and again by 0.78 points after recovery. This profile follows the joint $\rightarrow$ bottleneck $\rightarrow$ recovery sequence: bottleneck training routes image evidence through the decomposed latents during answer generation, whereas recovery restores direct image access. The intervention therefore captures stage-specific dependence on the latent intermediates induced by bottleneck training, rather than a permanent requirement after recovery.

We next test whether recovery preserves a usable latent-mediated visual route over each complete benchmark. All conditions use the same intervention-compatible attention path. In *latent-only*, non-latent queries cannot read image keys, while latent-token queries retain image access and downstream queries retain latent-token access. In *neither*, downstream queries additionally cannot read latent-token keys. Their paired difference therefore isolates the image $\rightarrow$ latent $\rightarrow$ downstream route without requiring it to be the model’s only route under ordinary full attention. Table 3(b) reports the corresponding full-benchmark accuracies in the main text.

GeoLatent’s latent-only route retains 71.3%, 71.1%, and 77.4% of its full-attention score and outperforms the corresponding neither condition by 5.31, 3.72, and 1.76 points on SPAR-Bench, SPBench, and ViewSpatial, with clustered 95% CIs [1.81, 9.03], [2.16, 5.31], and [0.53, 5.98]. The direct-recovery control gains approximately zero. The curriculum difference-in-differences estimates are 5.37 points (95% CI [1.78, 9.10], $p=0.0313$), 4.06 points (95% CI [2.09, 5.96], $p=0.00030$), and 1.78 points (95% CI [0.47, 6.33], plus-one $p=1/20001$), respectively. Resampling and sign flips use the predefined task clusters for SPAR-Bench and scene clusters for SPBench/ViewSpatial. Thus bottleneck training establishes a latent-mediated capability that remains available after recovery, even though ordinary full-attention inference also has a restored direct visual route.

Together, the diagnostics separate stage-specific evidence. Rank and assignment statistics measure global GEO organization, while POS/DIR decoding measures local geometric decodability. At the bottleneck checkpoint, edge interventions show use of the latent intermediates; after recovery, the decomposed representation remains structured while direct image access is restored.

B.9 CONTROLLED NO-GEO CAPABILITY PROFILE

The same-data no-GEO control follows the full curriculum on all 105,928 samples while removing the GEO span and its VGGT supervision. Its 62.4% three-benchmark mean falls to 40.3/38.1% under blank/mismatched images, confirming substantial dependence on visual input. GeoLatent reaches 62.8% while additionally providing a differentiated GEO span ($r_{\text{eff}}=3.62$, $I_{\text{assign}}=0.071$), stronger ViewSpatial visual reliance (VEF 88.3% versus 60.1%), and a 5.9-point camera-relative-direction gain. Person-centered direction additionally requires observer orientation R_{person} , which the current camera-frame vocabulary does not encode and which motivates future extensions.